Sales Prediction using Python Project Report

Objective:

The objective of this project is to predict product sales based on advertising expenditure across different marketing platforms such as TV, Radio, and Newspaper. Machine Learning techniques are used to analyze the relationship between advertising investments and sales performance.

Dataset Description:

The dataset contains:

- TV Advertising Budget
- Radio Advertising Budget
- Newspaper Advertising Budget
- Sales Revenue

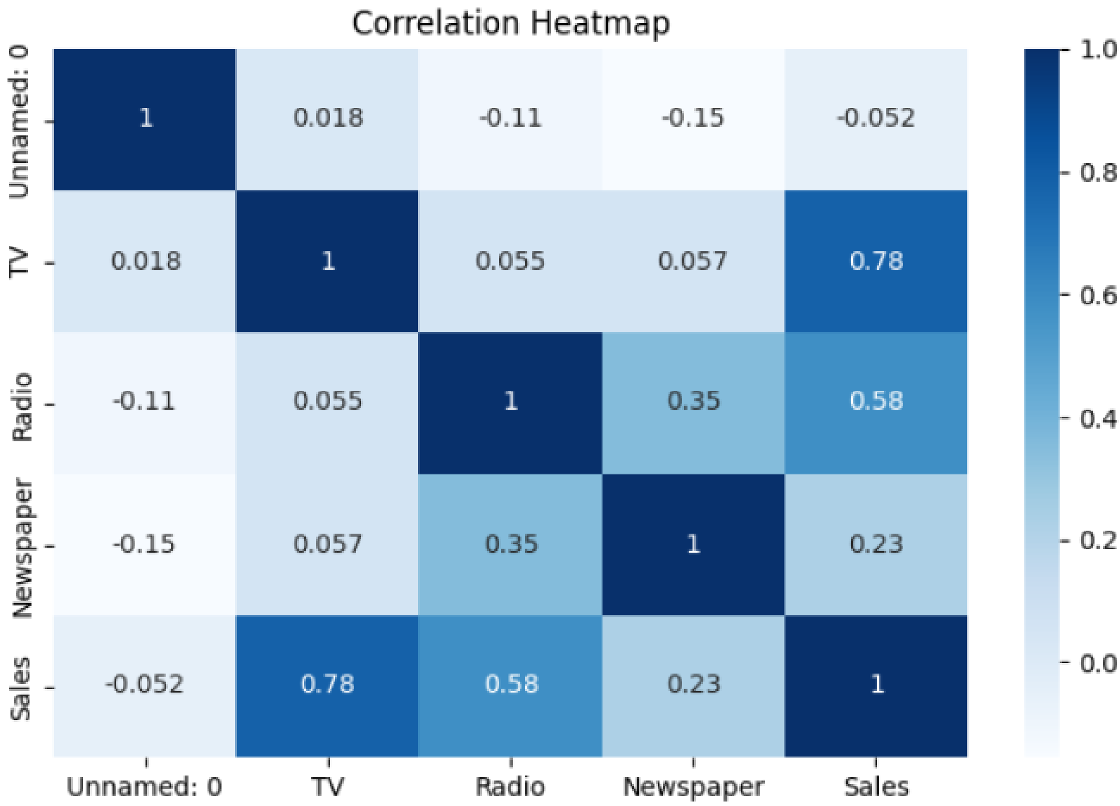
Technologies Used:

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-Learn

Data Cleaning:

- Checked for missing values
- Verified data types
- Prepared data for model training

Correlation Analysis



Observation

TV advertising shows the strongest positive relationship with sales. Radio advertising also contributes significantly, while Newspaper advertising has comparatively less impact.

Machine Learning Model:

Model Used

Linear Regression

Workflow

Data Collection

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Data Cleaning

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Feature Selection

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Train-Test Split

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Linear Regression

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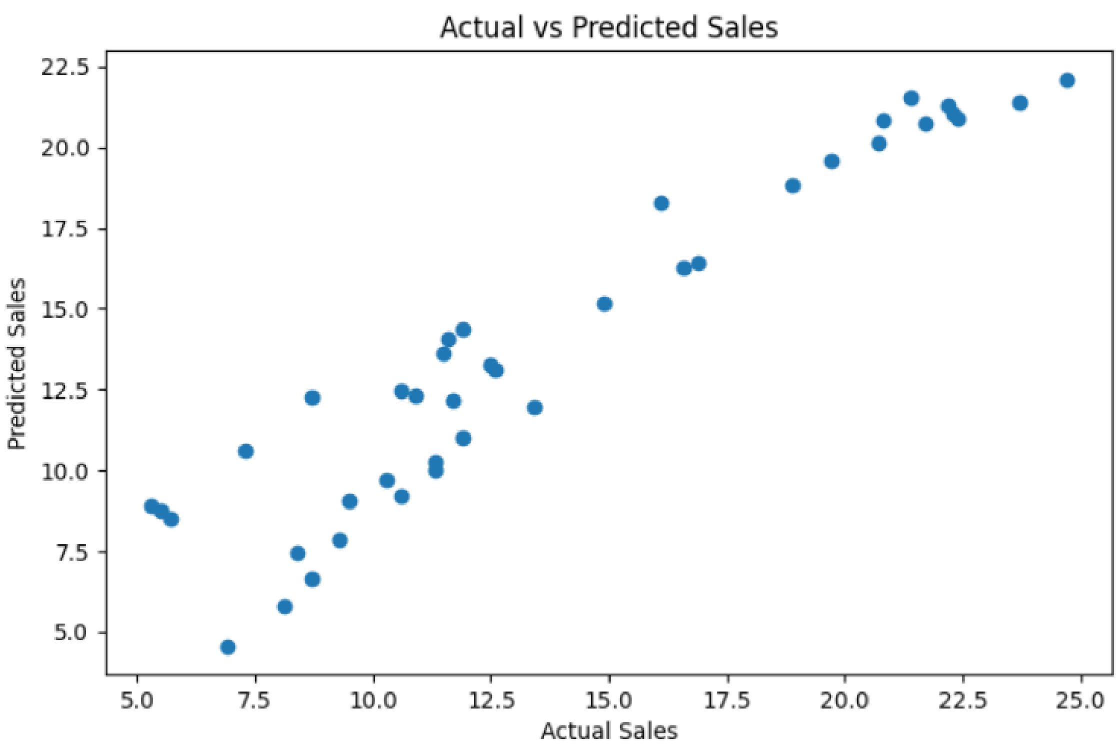
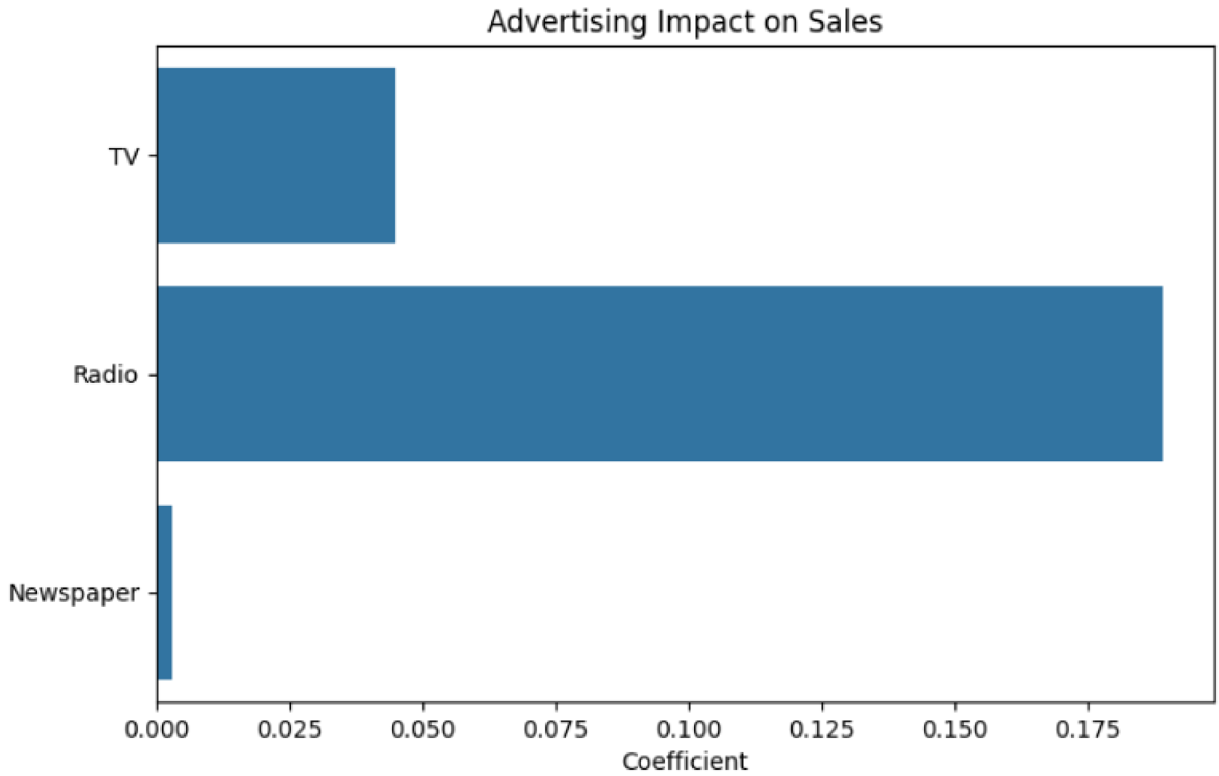
Sales Prediction



Observation

Predicted sales values closely follow actual sales values, indicating good model performance.

Advertising Impact Analysis



Findings

- TV advertising has the highest impact on sales.
- Radio advertising contributes positively to revenue growth.
- Newspaper advertising has the least influence.
- Businesses should focus marketing budgets on TV and Radio channels.

Key Insights

1. Advertising expenditure directly affects sales performance.
2. TV advertising is the most effective channel.
3. Radio advertising provides additional sales growth.
4. Newspaper advertising has limited influence.
5. The Linear Regression model successfully predicts sales based on advertising budgets.

Business Recommendations

- Increase investment in TV advertising.
- Maintain strategic Radio campaigns.
- Optimize Newspaper advertising spending.
- Use predictive analytics for future marketing decisions.